

Genebanks Accelerator

Inception Report

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List of acronyms

- **AoW** – Area of Work
- **ARIs** – Advanced Research Institutes
- **AWS** – Amazon Web Services
- **B4T** – Breeding for Tomorrow
- **CBD** – Convention on Biological Diversity
- **CGN** – Center for Genetic Resources, University of Wageningen, Netherlands
- **CGRFA** – Commission on Genetic Resources for Food and Agriculture
- **CoP** – Community of Practice
- **CWANA** – Central and West Asia and North Africa
- **DSI** – Digital Sequence Information
- **EA** – Evaluability Assessment
- **ESA** – Eastern and Southern Africa
- **GH/GHU** – Germplasm Health Unit
- **GGCE** – GRIN Global Community Edition
- **GRIN** – Germplasm Resources Information Network
- **HLO** – High-Level Output
- **IAES** – Independent Advisory and Evaluation Services
- **ISDC** – Independent Science for Development Council
- **IPK** – Leibniz Institute of Plant Genetics and Crop Plant Research
- **ISTA** – International Seed Testing Association
- **ITPGRFA** – International Treaty on Plant Genetic Resources for Food and Agriculture
- **LAC** – Latin America and the Caribbean
- **LPAs** – Long-Term Partnership Agreements
- **LTGs** – Long-Term Partnership Grants
- **MELIA** – Monitoring, Evaluation, Learning and Impact Assessment
- **MLS** – Multilateral System (of the Plant Treaty)
- **NARS** – National Agricultural Research Systems
- **NPPOs** – National Plant Protection Organizations
- **NUS** – Neglected and Underutilized Species
- **ORT** – Online Reporting Tool
- **PCU** – Programme Coordination Unit
- **PDCI** – Passport Data Completeness Index
- **PGRFA** – Plant Genetic Resources for Food and Agriculture
- **PORB** – Plan of Results and Budget
- **PPU** – Portfolio Performance Unit
- **QMS** – Quality Management System
- **RPPOs** – Regional Plant Protection Organizations
- **SMTA** – Standard Material Transfer Agreement
- **SOPs** – Standard Operating Procedures
- **SQM** – Seed Quality Management
- **ToC** – Theory of Change
- **USDA** – United States Department of Agriculture
- **WCA** – West and Central Africa
- **W3** – Window 3 (of CGIAR funding) – Bilateral Agreements

Executive summary

The Genebanks Accelerator plays a foundational role in CGIAR's 2025–2030 Portfolio, safeguarding and enabling access to the world's most diverse collections of crop genetic resources. During the Inception Phase (September 2024–June 2025), the Program focused on translating its high-level vision into operational priorities while responding to funding constraints and aligning with partner and stakeholder feedback.

The high-level vision—to conserve and make available crop diversity in the most effective, equitable, and innovative ways—remains unchanged. However, practical adjustments were necessary. Due to delays and changes in the disbursement of core CGIAR funds as well as changes in the structure of the Accelerator (i.e., the proposal had been written for a Program), the Genebanks Accelerator concentrated its efforts on core Areas of Work (AoWs): **AoW 1 (Biodiversity Conservation)**, **AoW 3 (Genetic Resources Policy)**, and **AoW 4 (Germplasm Health)**. These areas ensure compliance with global standards, sustain legally mandated conservation services, and maintain CGIAR's role as a trusted global provider of germplasm.

In contrast, **AoW 2 (Strategic User Engagement)** and **AoW 5 (Strengthening capacity for in situ and ex situ conservation globally)** were considerably reduced and partially delayed, though significant groundwork was laid for scaling user-centric innovation, digital platform development, and regional partnerships.

A donor earmarked USD 1.26 million for AoW5, prompting a reallocation of USD 730,000 from other Areas of Work to match the investment and strengthen training and stakeholder engagement. Although necessary, the reallocation simply redirected cross-cutting activities and staff time already aligned with AoW5 goals. Since the earmark was announced at the end of the Inception Phase, there was limited time to adjust. Relevant activities from other AoWs were consolidated under the capacity strengthening component to align with donor requirements without altering the Accelerator's overall Work Plan. This reallocation ensures timely delivery of AoW5 while maintaining collaboration and coherence across all AoWs.

Key inception deliverables included:

- Completion of a full **comparative advantage analysis** demonstrating CGIAR's unique legal, technical, and institutional positioning in global genebank operations.
- Execution of a **prioritization exercise** to target impactful and operationally viable HLOs, and regions when applicable.
- Mapping of bilateral and W3 projects to align strategic use of these funds to complement core operations.
- Strengthened alignment with cross-portfolio programs and accelerators such as **Breeding for Tomorrow**, **Digital Transformation**, and **Capacity Sharing** among others.

While the lack of integration of **CIFOR-ICRAF** and **World Vegetable Center (WorldVeg)** genebanks, which had initially been requested by the CGIAR Independent Science for Development Council (ISDC) and Independent Advisory and Evaluation Services (IAES), represents a missed opportunity to broaden the Accelerator's crop and regional scope, the Inception Phase reaffirmed the genebanks' strategic direction and readiness to scale innovation, policy engagement, and global partnerships. As part of this strategy, the CGIAR genebanks will partner with the two centers through AoW5, which focuses on building the global system for conservation and use of plant genetic resources. The genebanks now move into full implementation with a clear roadmap and renewed momentum to deliver on CGIAR's biodiversity and food system goals.

1. Program/ Accelerator overview

1.1 High-level Vision

The high-level vision of the Genebanks Accelerator, as outlined in Section 2 of the proposal, remains unchanged. The Accelerator envisions a future where the unique crop diversity conserved by CGIAR Genebanks is preserved and made available in the most effective and efficient ways to support more resilient, diverse, and environmentally sound agriculture and landscapes. It responds to global megatrends and SDG goals such as biodiversity loss, and climate change. Over the next five years, the Genebanks Accelerator aims to be a global leader in enabling smarter use of crop diversity, strengthening partnerships, and utilizing advanced technologies such as Artificial Intelligence (AI) and genomics to enhance accessibility and impact.

While the vision remains unchanged, the Inception Phase required practical adaptation. Due to delays in budget disbursements to CGIAR Centers, implementation focused on core functions under Areas of Work (AoWs) 1 (Biodiversity Conservation), 3 (Genetic Resources Policy), and 4 (Germplasm Health). These AoWs cover foundational operations like germplasm conservation and distribution, compliance with phytosanitary and access and benefit-sharing regulations, and engagement in mission-critical international policy making processes. Meanwhile, internal exercises like comparative advantage analysis and prioritization were conducted, particularly for AoWs 2 (Strategic User Engagement) and 5 (Strengthening capacity for in situ and ex situ conservation globally), which are essential but required further planning. These shifts reflect a flexible, responsive approach to maintaining program integrity while navigating funding constraints.

The 2025–2030 Genebanks scope (see page 14 of the proposal) included CIFOR-ICRAF and WorldVeg. Despite the clear rationale and endorsement—including from ISDC and the Independent Advisory and Evaluation Service—for expanding the Accelerator to these two international genebanks, resource constraints during the Inception Phase meant they could not be included in key activities as part of the Accelerator. This was decided on the basis of an internal costing exercise which estimated the costs of including these two Genebanks into the CG was around 2M USD on top of the regular funding, which was at the end not allocated to the Accelerator, that received only around same budget of 2024 when these two Genebanks costs were not considered. We have considered including these two genebanks as a priority in a “surge scenario”. This proposal was submitted to the CG which ended up not being approved due to organization-wide cost restructuring.

As a result of the above decisions:

- AoW 1 (Biodiversity Conservation) was unable to support CIFOR-ICRAF and WorldVeg in their routine operations.
- AoW 2 (Strategic User Engagement) could not extend outreach, product and tools development, or data integration efforts to the unique users and crops serviced by these genebanks, limiting global reach and user diversity.
- AoW 4 (Germplasm Health) faced challenges incorporating the germplasm of these genebanks into harmonized health testing and phytosanitary frameworks.
- AoW 5 (Strengthening capacity for in situ and ex situ conservation globally) lost the opportunity to leverage these Centers’ extensive regional networks for training and knowledge sharing, to improve the global system of PGRFA conservation and use overall. Although those national partners can still be part of the communities of practice and networks currently led by CGIAR Genebanks, without more direct support from the Accelerator, their integration and contributions to the global system was sub-optimal.

1.2 Achievements from 2022—2024

- 1- **Conservation of Plant Genetic Resources:** Five CGIAR Centers (AfricaRice, Alliance of Bioversity and CIAT, ICRISAT, IITA, and IRRI) manage seed collections that meet performance targets for availability, safety duplication, documentation, and quality management, as established and audited by the Crop Trust, as a pre-requisite to sign Long-Term Partnership Agreements that will allow partial funding in perpetuity to the Genebank. CIMMYT wheat is expected to reach these targets by 2025 and CIP potato (the first clonal collection) in 2026. Improvements implemented due to a strong quality management system (QMS) laid the groundwork for a single Genebank Process Model to cover all operational steps in seed, tissue culture, and field collections.
- 2- **Global Reach and Compliance:** From 2022 to 2024, the genebanks distributed more than 300,000¹ germplasm samples to 123 countries. About half went to users outside CGIAR, primarily to advanced research institutes, universities, and national breeding programs in low-income countries. All distributions underwent rigorous controls in Germplasm Health Units (GHUs) and were compliant with national and international laws and policies governing access and benefit-sharing.
- 3- **Technology and Innovation Integration:** The Genebanks Initiative pioneered the use of cutting-edge technologies to enhance conservation and increase the relevance of collections to users. This included using genomics to identify desirable traits, artificial intelligence to screen samples for resilience to stresses, and cryopreservation for crops that cannot be conserved as seed, successfully implemented at CIP and Alliance of Bioversity and CIAT genebanks. AfricaRice led work to establish a protocol for the automated characterization of rice grains using a multispectral videometer. IRRI led the work to harness AI-based methodologies in genebank operations and data management. Work is underway to develop a unified AI-enabled digital portal to provide seamless information on the diversity held across all CGIAR genebanks and make distribution more efficient and user-friendly.
- 4- **Strategic Capacity Sharing:** The CGIAR Genebank Initiative's Work Package 4 (WKP4) achieved significant progress in strengthening partnerships and building capacity among National Agricultural Research Systems (NARS), particularly in low- and middle-income countries. Through regional workshops and the Communities of Practice (CoPs) in Africa, Asia-Pacific, and Latin America, CGIAR Centers collaborated with national partners to improve genebank operations, seed processing, characterization and evaluation, data curation, and conservation techniques such as tissue culture and cryopreservation. Workshops in Africa and Asia-Pacific brought together genebank managers from over 30 countries to harmonize standard operating procedures, establish quality management systems, and foster stronger data and policy integration. CGIAR Centers further supported NARS by training graduate students, generating high-density genotyping data, facilitating gap analyses, and promoting community engagement, including citizen science and producer-focused initiatives. Two targeted calls for proposals, developed with the Plant Treaty Secretariat, provided funding to 15 countries to address policy challenges and upgrade national genebank operations. These efforts were complemented by online training modules developed in partnership with the UK Open University, on genetic resources policy and seed quality management offering accessible learning platforms for national staff. Collectively,

¹ https://www.cgiar.org/flipbook-page/?pdf=https://storage.googleapis.com/cgiarorg/2025/05/INIT03_Genebanks_ATR.pdf

these activities enhanced collaboration, trust, and the effective functioning of the global genetic resources system.

- 5- **Policy advancements:** In November 2024, the CBD COP16 in Cali, Colombia, adopted Decision 16/2, establishing a multilateral mechanism for benefit-sharing from the use of Digital Sequence Information (DSI) on genetic resources². This mechanism, including a global fund called the Cali Fund, aims to fairly and equitably share benefits, including non-monetary benefits like capacity building and technical cooperation, with those who provide genetic resources. The reports prepared by CGIAR Genebanks team and its strategic participation in the deliberations were instrumental in ensuring the needs of genebank users were included in the newly established mechanism. This is reflected in a statement by CBD's Executive Secretary, Astrid Schomaker, that, "The negotiations that led to the operationalization of the Multilateral Mechanism on DSI, including the Cali Fund, by COP 16 were supported by a broad base of multistakeholder engagement. CGIAR has conveyed the plural perspectives of agricultural research eloquently and made a remarkable contribution to forging consensus in Cali at the most vibrant and inclusive COP in the history of the Convention."

1.3 Description of intended research

The Genebanks Accelerator provides the global agricultural research community with secure and long-term access to the world's most diverse, scientifically curated crop collections. CGIAR genebanks operate under an international legal framework, providing equitable and open access to genetic resources for all. The Genebanks Accelerator is committed to both excellence in routine operations and continuous innovation—ensuring that systems to conserve and make biodiversity available are smarter, faster, and easier to use by the improvement and development of new protocols for long-term conservation, exploiting AI-enabled platforms, user-focused research, and harmonized digital infrastructures. The target audience includes breeders, researchers, farmers, and students—any stakeholder working to adapt food systems to future challenges.

The Accelerator is dedicated to continual improvement: advancing the science of conservation itself (e.g. cryopreservation protocols, seed longevity research, phytosanitation), applying emerging technologies (e.g. artificial intelligence, data sciences and genomics), and developing supportive policies for more strategic, data-driven, and equitable use of crop diversity.

Most of the Accelerator goals are focused on assuring high international standards of conservation and availability of the genebank collection and its associated data and adhering to international policy and phytosanitary protocols. Generally, the Genebanks Accelerator does not conduct research-for-development in the conventional sense but dedicates efforts to improve its core activities and engage with partners and users. It therefore contributes indirectly and strategically enables agricultural research.

Genebanks teams collaborate on key research products with other Science Programs and Accelerators, for example on targeted trait discovery projects. These include the sourcing, genotyping, phenotyping, and selection of genetically diverse materials from germplasm collections. Such projects require co-developed protocols, standardized ontologies, and shared data systems. Some of the Accelerator's collaborations include Breeding for Tomorrow and Digital Transformation. Through these partnerships, Genebanks provide the critical resources that feed into the breeding and wider research pipeline.

² <https://cgspace.cgiar.org/items/0f9c2ca1-7d3d-41a9-8f69-d006e7642838>

In the domain of user engagement research, the genebanks are adopting a strategic, data-driven approach to understand and meet the needs of a diverse and evolving user base. Planned research will:

- Identify and segment user typologies, map their needs, and design digital interfaces and data products accordingly.
- Deploy AI-enabled analytics to transform how users explore and select germplasm.
- Centralize and integrate data tools, platforms, and service pipelines to create a seamless user experience from germplasm request to delivery.
- Develop products including subsets that promote use of diversity.

This research is guided by the following key questions:

- Who currently uses genebanks (both CGIAR and non-CGIAR), and who could benefit from them more because of appropriate outreach and tailored access?
- What types of information and tools would make collections more accessible and actionable for diverse user groups?
- How can CGIAR's genetic resource information be consolidated and made universally available?
- What innovative products—such as curated subsets, seed kits, or introgression lines—can be developed to directly meet user needs?

By addressing these questions, the Genebanks Accelerator will not only enhance its service model but also play a transformative role in ensuring crop diversity contributes meaningfully to global resilience and innovation.

Further information on key research is included in the Theory of Change section (Section 3.1) with specific research under select HLOs.

2. Codesign and partnerships

For most of the proposal design, no co-design element was foreseen since the activities relate mostly to CGIAR genebanks internal and routine operations.

The Accelerator's most collaboration-oriented Areas of Work: AoW2 and AoW5 are designed to engage users to improve outreach and distribution of germplasm as well as collaborating with national partners through regional entry points to expand the reach of the CGIAR genebanks and to strengthen the Global System on Plant Genetic Resources for Food and Agriculture (PGRFA) as envisaged by FAO. In addition to the co-design sessions already undertaken, including the Stakeholder Consultations during the proposal phase, before the activities for these AoWs are implemented, relevant partners will be invited to co-design expected results.

During the Inception Phase the following internal and external partners roles have been fulfilled for the purpose of co-designing the Accelerator's activities, objectives and outcomes.

External partners

- 1- **Crop Trust:** one of the genebanks' donors and partners, has been involved in the design of part of the Theory of Change relative to HLO 1.4 *Effective and AI-ready data management systems using harmonized data standards across Centers*. The Crop Trust is responsible for developing GRIN Global Community Edition (GGCE) which many of the CGIAR genebanks are implementing and which will facilitate the harmonization of data across all Genebanks. It was essential that they should be involved in the initial design of the output and its indicators.
- 2- **Aarhus University:** the collaboration with Aarhus University has been ongoing for several years. The focal point of the university has been involved in designing the indicators and activities under HLO 1.2. *Innovations to enhance the efficiency of conserving collections*, in particular, Aarhus University holds the role of guiding the Community of Practice on Seed Quality Management and, for this reason, they were instrumental in designing one of the online courses which is part of the activities under AoW5 and which supports AoW1.
- 3- **National partners (NARS)** have been thoroughly involved during the proposal phase through stakeholder consultation, individualized meetings, and feedback sessions following capacity building activities. During the Annual Genebanks Meeting held in September 2024, a session with NARS from the Southern Africa Development Community was organized to gather feedback on improving genebank services for NARS. This feedback was embedded in the HLOs for AoW 5. Through the existing communities of practices and networks, we continuously interact with partners and gather feedback from them on priorities for future engagement. We also engaged with partners during Science Week. National Genebanks are key stakeholders to a successful PGRFA Global System, and we do rely on strengthening our collaborations with them as part of the activities to be developed under AoW 5.
- 4- **The Plant Treaty Secretariat** partnered closely in envisaging/framing genetic resources policy work. Both the Plant Treaty Secretariat and National Plant Treaty Focal Points supported, and will continue to support, the design of AoW3 capacity development activities, specifically HLO 3.3 *Draft laws, policies and guidelines developed to implement the Plant Treaty to exchange germplasm*.

Internal partners

- 5- **Digital Transformation Accelerator:** in the context of developing HLO 2.3. *CGIAR-wide portal with AI-enabled digital infrastructure developed for germplasm data* and requests, consultation with the Digital Transformation Accelerator has been instrumental in identifying the needs of the genebanks and linking them with capabilities, particularly relating to the AI and Digital Transformation Accelerator Hub in Abu Dhabi and to the collaboration with Amazon Web Services (AWS). Initial discussions with Digital Transformation were instrumental in shaping the high-level features of the AI infrastructure.
- 6- **Breeding for Tomorrow Program (B4T):** collaboration with B4T has always been close, with the genebanks providing a key service to breeders across CGIAR and beyond (more information on the operationalization of this partnership is contained in Section 8). Feedback from B4T helped shape AoW2, particularly HLO 2.1, for which market intelligence shared by breeders is fundamental. The feedback loop on market intelligence for key accessions could be improved as currently B4T's involvement and advice has been provided mostly at the center level, since this is where the collaboration is most effective: directly between the genebank and the breeding programs of specific crops. During the inception phase Genebanks AoW 3 continued to participate actively in efforts convened by B4T to develop a harmonized approach across CGIAR to germplasm licensing. Genebanks will continue to be actively involved in the years to come.

A clear framework for collaboration is essential to ensure that all stakeholders are engaged and that their contributions are recognized. The Genebanks Accelerator will continue engaging with other Science Programs and Accelerators and will be launching a new survey to help design a strategy for improved engagement on key activities.

3. Theory of change and MELIA

3.1 *Theory of change*

The full Theory of Change (ToC) and AoW ToC narratives with documented assumptions are found here: <https://toc.mel.cgiar.org/toc/16685dd2-9da6-4a10-9270-dfdfb2dcdb53?relatedFlowId=39e86294-d092-4949-bb34-8857b4d5f021>

The Genebanks Accelerator aims to ensure the reliability of CGIAR conservation and genebank services now and into the future and to build a stronger global system of policies, mechanisms, institutions and conservationists making genetic diversity more widely available. The Genebanks Accelerator will be delivered through five interconnected Areas of Work (AoWs): AoW 1. Biodiversity Conservation, AoW 2: Strategic User Engagement, AoW 3: Genetic Resources Policy, AoW 4: Germplasm Health, AoW 5: Strengthening capacity for in situ and ex situ conservation globally.

Through AoW 1, CGIAR Genebanks will conserve and make available crop diversity in perpetuity with greater efficiency and effectiveness. AoW 1 supports the genebanks' operations and improves their related processes, collections, and facilities. CGIAR is uniquely placed to pilot and refine new conservation-related technologies across diverse locations and crops. AoW 1 supports genebanks in reaching and sustaining performance targets so that they become eligible for long-term partnerships with the Crop Trust. AoW 1 will maintain and create new partnerships with ARIs and universities, NARS and national genebanks, the Crop Trust, and the Plant Treaty Secretariat.

AoW 2 will enable researchers, breeders, students and other users to access diversity and associated data in a smarter and more targeted way for increased use. AoW 2 will implement a strategic approach to identify, prioritize, and characterize user typologies, analyze their needs, and proactively engage with user type representatives to determine how to improve the user experience in finding, selecting, and requesting germplasm from CGIAR genebanks. Enriched information, new products, and responsible AI innovation will help users identify materials and improve their experience ordering and obtaining germplasm, which is expected to facilitate breeding and research and potentially speed up their research. Improved user tools and outreach will simultaneously create more demand for genetic diversity and make it possible for users to target more precisely what germplasm from CGIAR hosted collections has the most potential to serve their needs. Successful user engagement may also generate opportunities to attract funding and sponsorship to support long-term conservation. AoW 2 will work with Breeding for Tomorrow, Digital Transformation, partner with NARS, ARIs and universities, and establish a user feedback loop.

AoW 3 will a) ensure CGIAR scientists comply with evolving international and national access and benefit sharing as materials move across international borders, b) contribute to ongoing international negotiations to make those laws more supportive of agriculture for development, and c) work with national partners to develop complementary national measures implementing those same laws. AoW 3 will focus primarily on the Plant Treaty's multilateral system of access and benefit sharing, making sure it evolves and is implemented in ways that are in harmony with the Convention on Biological Diversity's Nagoya Protocol on ABS and Decision 16/2 on digital sequence information. Partnerships in AoW 3 include the Secretariats of the Plant Treaty, CBD, CGRFA, National Plant Treaty Focal Points, national genebanks, DSI Scientific Network, Art 15 DGs and System Board, International genebanks, and Breeding for Tomorrow.

AoW 4 will ensure compliance with national and international quarantine laws as materials move across international borders. AoW 4 will develop and improve phytosanitary interventions to meet current and emerging phytosanitary demands, and identify, develop, and mainstream effective,

advanced phytosanitation and diagnostics technologies (genomic-based molecular methods, multispectral imaging, AI/ bioinformatics, etc.) for routine germplasm health monitoring. Partnerships in AoW 4 include Sustainable Farming and Breeding for Tomorrow, FAO-IPPC, NPPOs, RPPOs, Universities, ISTA, ARIs, and NARES. At the heart of AoW5 lies a bold commitment to strengthening national genebanks, with a particular focus on the global South, as an essential pillar of the global system for the conservation and use of PGRFA. The Crop Trust plays a pivotal global role in safeguarding PGRFA and has launched strategic initiatives to support national genebanks. Building on this foundation, CGIAR genebanks recognized for their robust technical expertise, operational excellence, and deep field experience, are uniquely positioned to establish long-term, transformative partnerships with NARS. To maximize impact, CGIAR genebanks and the Crop Trust will intensify their collaboration to empower NARS across the global South. Our strategy is anchored in a holistic capacity development approach that includes the creation of regional hubs, vibrant communities of practice, interactive online learning platforms, hands-on technical training, and tailored mentorship programs. Together, these efforts will build resilient systems that ensure the effective conservation and use of genetic resources at national and regional levels. While the principal focus remains on elevating the global south, the initiative maintains a global perspective. CGIAR genebanks may extend technical assistance to genebanks in the global North when needed. Moreover, strategic partnerships will be fostered with world-renowned genebanks, such as the Centre for Genetic Resources (CGN) in the Netherlands, the USDA genebanks in the United States, and the IPK genebank in Germany, who has cutting-edge expertise and excellent infrastructure for genetic resources conservation. These leading institutions will play an essential role in strengthening the global system by hosting NARS staff for immersive training programs and contributing to CGIAR led online and thematic courses. Through these synergies, AoW5 will help create a more interconnected, equitable, and resilient global network of genebanks capable of safeguarding biodiversity and ensuring its use for present and future generations.

Genebanks activities will contribute to CGIAR Impact Area (5) Environmental health and biodiversity and the Collective Target 11. By conserving and making available biodiversity, Genebanks are responding to SDG 2.5 and the Kunming-Montreal Global Biodiversity Framework Targets 4, 10, 13, and 20, and the FAO's Global Plan of Action on Plant Genetic Resources for Food and Agriculture (PGRFA) and giving thousands of users access to plant diversity sources. Genebanks will indirectly contribute towards other CGIAR Impact Areas (1), (2), (3) and (4). During the inception phase CIFOR-ICRAF and the World Vegetable Center were not included as part of the Accelerator at the same level as the other 10 CGIAR Genebanks since the budget allocated to the Genebanks is slightly less than what was allocated in 2024 making it impossible to accommodate the routine costs of two new international genebanks, especially at the same time when we adjusted routine costs through a new cost study conducted five to six years after the previous one in which 2024 budget allocation was based. Other changes to the ToC included revising the high-level outputs and outcomes, refining and formulating new assumptions, and responding to ISDC feedback. Previous evaluation findings have influenced the ToC in multiple ways including fully integrating the work on Plant Genetic Resources policy, both outward and inward facing, into the Genebank Accelerator (AoW 3), expanding policy-related capacity building efforts beyond genebanks to other relevant stakeholders in the CGIAR system (AoW 3 and AoW 5), seeking to adopt a wider definition of users when designing outreach programs (AoW 2), establish a working group to review IT needs of CGIAR genebanks (Data Management CoP in AoW 1), promoting more agile phytosanitary operations and capturing opportunities in systems approach for the transboundary movement of CGIAR germplasm (AoW 4). Genebanks met with members of the Evaluability Assessment (EA) team and PCU in Nairobi during Science Week followed by a virtual meeting with PCU's Monitoring and Evaluation Specialist who supported Genebanks during the inception phase by providing sound advice and feedback in updating the ToC and MEL

framework. The EA team provided a partial evaluability assessment using Domain A: Intervention Logic of the EA Framework which helped Genebanks in finalizing the inception report.

3.2 *Monitoring, Evaluation, Learning, and Impact Assessment (MELIA)*

The full MELIA plan and Results Framework are accessible via the relevant attachments.

Monitoring, evaluation, and learning

The genebanks will continue to use an online reporting tool (ORT), which was set up in 2014 to monitor the status of individual collections with respect to performance targets. Through the QMS adopted in AoW 1, genebank operations, data, and accession numbers are audited internally and externally and are subject to regular review by the Crop Trust. The audits ensure that international standards are complied with, and deviations and risks are documented and managed. FAO's monitoring of UN SDG 2.5, State of the World of PGRFA, and the Global Plan of Action, the Crop Trust's oversight of long-term partnership agreements and grants, CBD's Global Biodiversity Framework, and perhaps most importantly, CGIAR Centers' Article 15 Agreements with the Plant Treaty's Governing Body all require annual, biannual, or periodic submissions from CGIAR and help monitor outcomes of CGIAR genebanks in all five AoWs. In addition, individual accessions, their data, and the availability status of national and international genebanks are also recorded and regularly updated in the Crop Trust's online portal Genesys.

AoW and communities of practice for seed and clonal crops and data management will be coordinated by dedicated staff, who will be responsible for implementing work plans and collating information and feedback from each genebank and GHU. Progress on high-level outputs will be reported also in the ORT in a suitable format to comply with system-level annual reporting requirements. By nature, AoW 2 will involve an iterative process of identifying target user typologies to engage and develop user entry points into genebank data resources. The success of this AoW will become evident in trends in the use of the online portal and germplasm requests monitored under AoW 1. Genebanks will adopt a robust feedback mechanism to monitor rates of response to users' requests and their satisfaction. Work developed through other projects (W3/bilateral) will not be monitored through Genebanks. However, the results of such work will provide new data and information to aid the selection of genebank materials, which will become evident in trends in germplasm requests and the use of materials. Genebank Management team meets every month to share financial, technical, and management updates. Discussions held during the meeting support better interaction and co-learning among genebanks, policy, GHUs and management, promote better interaction, and proper follow up on achievements, guiding decision making and potential adjustments to the Accelerator. The Annual Genebank Meetings (AGMs) are an important forum to review and reflect on the CGIAR Genebanks past year's activities, progress and achievements, and to make plans for the next year. National partners, CGIAR programs, and technical and scientific experts are represented in the AGMs, and the location of the meeting is in a different region each year to assure better interaction with multiple stakeholders from different regions. By doing this, we are constantly consulting with partners in different regions of the world, understanding their realities and needs so we can better plan the priorities based on a face-to-face consultation with partners. Topics discussed at the AGMs are taken forward to the management team meeting for further discussion and action. From these consultations with multiple partners, we have developed the structure of the capacity sharing activities (such as webinars, workshops, and training courses) targeted within and outside genebanks.

Genebanks Accelerator is proposing the following MEL studies (more detailed in the MELIA attachment):

- "Assessing the application of best practices among national partners".

- “In situ and on farm site monitoring.”
- “Review the Scientific Impact of the SQM and QMS Communities of Practice”.
- “Study on the Use of Germplasm Requested from CGIAR Genebanks”.

Impact assessment

The impact of genebanks is notoriously challenging to assess given the lack of germplasm tracking, obscure pedigrees of improved materials, complex attribution patterns along a long impact pathway, and the wide range of users and purposes to which genebank materials are put. Much has been published on the economic impact of traits or diversity sourced from genebanks, and some cost-benefit analyses have concluded that the costs of long-term conservation will be minor compared to the scale of impact and returns on investment of just one disease-resistant trait coming from screening a genebank collection in a major crop. One of the most impressive impacts occur where “black swan” events (e.g. UG99 wheat stem rust, banana Fusarium tropical race 4) demand the screening and use of diversity found nowhere else but genebanks.

Opportunities will be taken to participate in large, rigorous research undertaken by downstream partners and Science Programs to which genebanks contribute. For instance, Genebanks may benefit from adoption research undertaken by Breeding for Tomorrow or other investigations from different breeding programs. AoW 2 will improve germplasm tracking, its use, and follow up with users to monitor potential outcomes of ordering genebank materials. The increasing use of common standard DOIs by CGIAR and other breeders would considerably improve tracking the contribution of genebank materials to widely adopted improved cultivars. Impact stories have been written by postgraduate students and have highlighted the contributions of genebanks to cultivars growing in farmers’ fields. While these published investigations are limited in depth and scope, they suit the scope of genebanks as service providers and the wide range of impacts to which they contribute. AoW 4 will produce an Impact Assessment study with the tentative title “From Exchange to Impact: How GHUs Catalyze Genebanks-Breeding Germplasm Exchange for Invasive Cassava Disease Management” in collaboration with Breeding for Tomorrow, Sustainable Farming, IITA, and partners. Collaboration on MEL investigations with W3/bilateral projects will be explored during phase 2 of the mapping exercise.

4. Comparative advantage analysis

The Comparative Advantage exercise highlighted that CGIAR Genebanks are very well positioned in the global ecosystem for crop biodiversity conservation. During the proposal phase, the ISDC suggested that carrying out this kind of exercise for the Genebanks was very difficult given the nature of their work. This was the ISDC's feedback: *[CGIAR] is the only institution with the institutional (Treaty), physical, and human capitals (including the GHUs), and incentive structure required to provide this deliverable. In fact, it is the only global source of genebank materials that is mandated to provide these materials to all eligible users (demand partners).*³

With their unique legal mandate and positioning, the CGIAR Genebanks have a strong advantage compared to other actors in this field, some of which are positioned only to undertake key activities partially, often due to lack of means or network, or mandate. While some national genebanks are well managed, and certainly many of them conserve crop species that are not included in CGIAR genebanks, many lack adequate resources in continuity to operate on a routine basis (e.g. to undertake viability monitoring, regeneration and distribution, prevent backlogs of materials in processing) at the level required to respond to demands to distribute globally. The CGIAR genebanks, thanks to dedicated quality management activities under the guidance of the Crop Trust, have obtained a leading position in regard to reliable conservation and creating access to PGR, and thus created a model for genebank operation that can be used by all other actors in this field.

The role of national genebanks is conceptually different from international genebanks. Although they can respond to international requests, they are often mostly focused on their own country's needs. Even then, national germplasm distribution is often limited. This is evidenced by multiple actors, including national genebanks, from the same country, requesting the same germplasm from CGIAR Genebanks. CGIAR genebanks and breeding programs are responsible for nearly 90% of plant genetic resources exchanges under the Plant Treaty's multilateral system of access and benefit sharing. CGIAR PGRFA collections are not owned by any nation or organization. Since 1994, pursuant to the agreement Centers signed with the FAO, CGIAR Centers hold their PGRFA collections 'in trust for the international community'. This sets the CGIAR Centers and their genebanks apart from national genebanks whose mandates are mainly domestically focussed, and as a result do not provide much PGRFA beyond their own borders (subject to some important exceptions, e.g. USDA, NordGen, CGN, IPK). The Plant Treaty states that "Contracting Parties recognize the importance to this Treaty of the ex-situ collections of plant genetic resources for food and agriculture (PGRFA) held in trust by the International Agricultural Research Centres (IARCs) of the Consultative Group on International Agricultural Research (CGIAR)" and called upon Centers to sign agreements with the Governing Body to manage the collections under the Plant Treaty framework. The 11 Centers hosting international in-trust collections signed such agreements in 2006 and CGIAR has operated under the Plant Treaty framework ever since, sending over 6 million samples under more than 60,000 SMTAs to recipients around the world. CGIAR Genebanks contribute more than 50% of the data on accessions available for distribution in Genesys, the online portal of genebank accession data, with a very high passport data completeness index (PDCI). Thirty-four countries outside Europe have provided data on Genesys. The majority of national genebanks are yet to confirm which of the PGRFA they will make available under the Plant Treaty's multilateral system of access and benefit-sharing, and very few have shared passport and characterisation data about their collection, leaving those collections with much lower PDCI when compared to the CGIAR genebanks data available in this portal.

³ Independent Science for Development Council. 2022. Identifying and Using CGIAR's Comparative Advantage. Rome: CGIAR Independent Advisory and Evaluation Service <https://cgspace.cgiar.org/server/api/core/bitstreams/45e8a519-b408-4074-894e-a8cb347a23e5/content>

All of the above exemplifies the fact that, especially in the context of Area of Work 1, CGIAR genebanks have a clear comparative advantage due to effective quality management, and specialist skills and networks at the international level. CGIAR genebanks are trusted partners to many NARS, they disseminate new knowledge and often host safety backups for partners. The Comparative Advantage exercise confirmed that maintaining and enhancing international Genebanks is a core function that aligns tightly with CGIAR's strengths and legal mandates. It also confirms that many national partners rely on the CGIAR Genebanks as a source of new knowledge and materials with very high standards in terms of seed quality and data associated with them. Furthermore, future generations of users will be able to breed their new varieties and thus feed the world on the basis of the PGR now conserved in the CGIAR genebanks.

In the proposal, the genebanks set out the goal to conclude an in-depth comparative advantage analysis for Area of Work 2: Strategic user engagement (see page 9 of the [proposal](#)). During the Inception Phase, the Comparative Advantage for Area of Work 2 revealed that no party (outside of the CGIAR) can undertake similar work at the same rate, efficiency and with the same global impact as CGIAR Genebanks. Nonetheless, the Accelerator recognizes the importance of working with other Programs and Accelerators, such as Digital Transformation, in the implementation of certain High-Level Outputs. This will be illustrated more in depth in Section 8 – Cross Portfolio Linkages.

For Areas of Work 3 and 5, several actors were identified as technically capable of undertaking aspects of the Genebanks Accelerator's work, but they face limitations that prevent them from doing so as effectively as CGIAR. These actors include FAO, national research institutes, universities, and regional genebanks. While they may possess relevant knowledge or infrastructure, they often lack the global mandate, intensity, coverage, and integration that CGIAR provides. For instance, FAO cannot match CGIAR's operational depth; national research institutes and universities may have technical expertise but are not structured to deliver long-term or large-scale conservation or to provide an international perspective; and regional genebanks, though knowledgeable, typically play a supporting or training role, rather than one of global custodianship. These limitations reinforce CGIAR's unique positioning and justify its continued leadership in this domain. As far as ensuring that CGIAR operations comply with international rules and standards, it makes sense that CGIAR should take the lead in providing capacities and collaboration across actors involved in PGRFA conservation and its correlated aspects such as policy and germplasm health, which will ultimately lead to a stronger PGRFA Global System where different actors can play their roles at the highest level, but that cannot combine the leadership needed and the technical expertise. It is important to highlight, however, that the CGIAR Genebanks will play this role in collaboration with other key partners such as the Crop Trust and the Plant Treaty.

As far as engagement in international negotiations is concerned, it is important to note that the international rule-making that the genebanks want to influence mostly takes place under the auspices of FAO and the Plant Treaty. As stakeholders active in the field of research and development, the genebanks' perspective, and role during negotiations is necessarily very different from that of FAO and the Plant Treaty Secretariat.

The complementary nature of FAO's Commission on Genetic Resources and its working groups, the Plant Treaty Secretariat's and the genebanks' respective contributions for supporting national level capacity strengthening is reflected in the fact that FAO, the Plant Treaty Secretariat and Bioversity International (then IPGRI) created a Joint Capacity Strengthening Programme to help countries implement the Plant Treaty's multilateral system of access and benefit sharing. Every biennial meeting of the Plant Treaty's Governing Body (now comprising 153 member states) over the last 10 years has welcomed the work of the Joint Programme and encouraged donors to support its continued work. This well-established programme and track record underscore our

comparative advantage for promoting HLO 3.3 *Draft laws, policies and guidelines developed to implement the Plant Treaty to exchange germplasm.*

The consistency in high ratings for Comparative Advantage for AoW 4 reaffirms CGIAR's strong positioning to lead in the development and application of phytosanitary practices essential for global germplasm exchange and conservation. GHUs were established due to weak phytosanitary capacity in the regions of CGIAR operations and to serve as a 'single gateway' for the phytosanitary safe exchange of germplasm and breeding programs to ensure compliance with the national and international phytosanitary regulations. GHUs work in close collaboration with NPPOs. CGIAR GHUs have kept pace with the changing pest and regulatory issues and possess the specific knowledge, standard operating procedures (SOPs), and partnerships needed to develop comprehensive systems to address new and emerging phytosanitary risks. GHUs also contribute to strengthening national and regional phytosanitary capacity by serving as diagnostic hubs for supporting surveillance and capacity development activities.

In the context of AoW 5 (Strengthening capacity for in situ and ex situ conservation globally), a few actors that could undertake similar work emerged in the context of all HLOs. For example, for HLO 5.2 *Communities of Practices and networks for diversity/gap analyses of crops at a regional level for strategic conservation and use*, there are several regional organizations that could be considered having similar scope as CGIAR Genebanks. It is important to assert that most regional actors, for example APAARI in Asia, have a mandate focused on knowledge management and sharing international standards and techniques for the development of sustainable agriculture in Asia, which includes as part of the scope biodiversity conservation and agrobiodiversity, while CG Genebanks set the international standards for saving crop diversity at risk and can play a strong role in collaborating with such actors. Additionally, universities and regional actors play a role in establishing Communities of Practice locally, nevertheless, some of these institutions could be technically positioned to undertake this HLO but not politically positioned. This is the case for many of the national genebanks in the global south as well, as often the costs of implementing long-term conservation are high and the constant insufficient funding for conservation whether from national governments or from other scarce sources, make it more complex to achieve long-term goals. Another major setback of many national genebanks is the high rotation of staff, which makes it even harder to keep such networks working with the necessary continuity. The combination of relatively stable funding with continuity of the genebanks' core work, global scope and already implemented networks allow the CGIAR Genebanks to provide partners with unique support on specialized topics.

Completing the Comparative Advantage exercise goes directly in response to another ISDC comments for feedback criteria 3, stating that "the claim that CGIAR holds a unique position in providing genetic resources is questionable", A more elaborate response to this item can be found in Appendix I.

Please refer to the Accelerator's full Comparative Advantage exercise in the attachment for more information.

5. Prioritization

5.1 Knowledge-driven priorities

The genebanks' prioritization exercise highlighted a need for an Accelerator-specific approach given the several characteristics of the Accelerator's budget and activities differ from other Programmes and Accelerators. Three of the five areas of work (AoW 1, AoW 3 and AoW 4) are already considered priorities for the Genebanks Accelerator and for CGIAR since they are part of CGIAR's responsibility to the international community and must be prioritized in any circumstance. These are core activities of the genebanks and are subject to a budget ring fencing mechanism, approved by the System Council, meaning they should not be deprioritized in any way. Without them, CGIAR would fall short of its legal obligation to manage long-term biodiversity conservation globally.

Due to CGIAR's legal obligations and to the recognized challenge of securing funding for conservation activities, Genebanks have been identified as a pooled funding priority by CGIAR funders since 2012 and more recently by CGIAR's Global Science Team (GST). The Genebanks also hold an agreement with the Global Crop Diversity Trust, which, through an Endowment Fund, partially covers the Accelerator's essential operations in perpetuity.

The above explains why the Genebanks Accelerator could not include AoWs 1,3, and 4 in the prioritization exercise. This decision was made in agreement with the Portfolio Performance Unit and following deliberations of the Accelerator Management Team.

The prioritization exercise therefore only covered AoW 2 (Strategic User Engagement) and AoW 5 (Strengthening capacity for in situ and ex situ conservation globally). Regional prioritization was filled out for AoW 5 only, since activities listed under AoW 2 are not region-specific. AoW 2 High Level Outputs have been prioritized against each other.

5.2 Final decisions on the Program/Accelerator's direction and design

Following the prioritization exercise, several key points emerged:

- The work of the Genebanks is mostly relevant to Impact Area 5. *Environmental Health and Biodiversity*, to which Genebanks contribute directly. Other key Impact areas are 1. *Food Security* and 4. *Climate Adaptation*. Nonetheless, while the Genebanks Accelerator is conducting activities with these Impact Areas in mind, its first objective is to provide a service, which only indirectly achieves progress towards these Impact Areas.
- For AoW 2, instead of differentiating by region, we have used the scoring process to differentiate these global HLOs by their relative contribution to different Impact Areas and their comparative importance within the AoW, based on their anticipated value-add to user experience, data quality, or strategic foresight. HLOs 2.1 through 2.4 represent global-level digital infrastructure and strategic planning activities, which are designed and implemented centrally through CGIAR genebanks. Their outputs — such as digital platforms and AI-enabled data tools — are intended to serve users across all regions equally once deployed. As such, these HLOs do not have region-specific differentiation in impact potential or delivery logic. The regional context does not influence their implementation pathway or expected outcomes. Therefore, the same score was applied across all CGIAR regions.
- For AoW 5, the exercise showed that the geographical areas that emerged as areas where the highest impact is possible across all Impact Areas, are South Asia and Southeast Asia. While genebanks are present in these regions, they are not currently being prioritized. This is due to

several reasons including budget, lack of NARS participation and need, presence of stronger regional actors, and the enabling environment overall. A clear example of this is the South Asia region. As a standalone measurement, the average impact score, for example, of Impact Area 1 – Food Security is roughly 0.86. When this is weighed against its enabling environment the impact across all genebanks activities decreases massively, to roughly 0.46. Therefore, in the case of the genebanks, the decision was made to continue investing in lower scoring but more operationally viable regions (LAC, CWANA, WCA and ESA across all Impact Areas). This is a strategic trade-off that maximizes actual impact as the activities are already initiated, it also follows the recent [ISDC recommendations](#) which stated “consider collaboration with national / international genebanks that also comply with ITPGRFA where they have capacity to preserve species and direct CGIAR efforts to under-resourced regions for capacity development such as East and sub-Saharan Africa that are diversity hotspots”.

- While the work of the Genebanks Accelerator is not directly connected to Impact Areas 2. *Poverty Reduction* and 3. *Social Inclusion*, some activities of the Accelerator could be considered as indirectly influencing progress of these areas of work, for example those related to High Level Outputs 5.2 and 5.3 on Communities of Practice and Strengthening Capacity as well as routine activities under HLO 1.1. As stated in the [proposal](#), CGIAR genebanks distribute a few thousand samples to farming communities, representing around 8% of their total annual germplasm distributions. Certain crops, cultivars, and traits (e.g. clonal crops such as potato and banana, forages, tree species) follow direct distribution routes more frequently than other crops depending on the (lack of) breeding activities in these crops, but also on the outreach of genebanks to different user typologies. This could prove to have an indirect impact towards Poverty Reduction and Social Inclusion. However, there are clear evident gaps for these impact areas which have not been addressed because they do not constitute a priority for the Accelerator currently. AoW 2 (Strategic User Engagement) could potentially promote approaches to specifically target the participation of women, youth, Indigenous People, and marginalized communities in accessing genetic resources and genebanks that are relevant to them. Nevertheless, this remains a secondary priority to the Genebanks.

Based on all the above, the Genebanks team, for the AoWs illustrated above, decided to continue focusing their work on the geographic areas where the Accelerator is already very active: LAC, CWANA, WCA and ESA and directly demanded by partners. While Step B suggests lower potential for scaled impact in these regions, when the enabling environment is introduced, these regions report high scores of potential impacts across most of the genebanks’ HLOs. In these regions, the genebanks have already established regional networks which helps with implementing activities and bringing positive impact. According to genebank staff experience, highly cooperative national partners in these areas significantly improve the enabling environment and likelihood of real, sustained outcomes.

While some high-score regions like South Asia and Southeast Asia may have greater potential impact on paper, success is less feasible due to institutional constraints, geographical isolation and willingness of national partners to interact. This does not mean that these regions will be excluded from current activities, but the prioritization exercise served as a basis for discussion on conscious investment, also considering the recent budget readjustments that were required across CGIAR. South Asia and Southeast Asia remain key regions rich in crop diversity, and collaboration with international, regional, and national partners will continue to ensure that diversity is safeguarded, and national partners are empowered. In case of a surge in budget, activities will be prioritized in these regions.

Exercises such as the W3 mapping and the Comparative Advantage analysis substantiated the decisions made during the prioritization exercise, showing that bilateral projects with national entities in the regions where the genebanks are most established tend to provide more data, outputs, and sustainable results compared to others. Further bilateral funding in South Asia and Southeast Asia would be extremely beneficial to reach desired results in these regions. Active participation from NARS not only contributes to the success of joint initiatives but also fosters long-term partnerships that are essential for building a resilient and integrated global genetic resources system.

Please refer to the Prioritization attachment for more information.

6. Alignment of W3 and bilaterally funded work

The Genebanks Accelerator is involved in a variety of bilateral agreements that complement its core functions.

Bilateral projects funded by the Crop Trust and FAO/NORAD directly support AoW 1, primarily by contributing to its main goal of sustaining conservation operations and securing long-term funding.

All Long-Term Partnership Grants (LTG), and Long-Term Partnership Agreements (LPA) financed by the Crop Trust contribute directly to the long-term financial sustainability of genebank operations, although the total funding provided currently is equivalent to ~30 percent of the total budget needed to effectively maintain the collections alive and available (essential core operations).

The LPA mechanism is the one used to partially fund the essential operations of international genebanks in perpetuity. The purpose of this agreement is to support long-term conservation and sustainable use of plant genetic resources through an endowment fund. These funds are granted to genebanks which meet performance targets based on international standards for biodiversity conservation, the genebanks must constantly meet these standards and report them to the Crop Trust. The LTG mechanism is a contribution that the Crop Trust makes to support genebanks reach these performance targets. They are much smaller than the LPAs.

Further outreach is required to access more bilateral funds for AoW 4 and ensure sustainability and complementarity of results between W1/2 and W3 funding. Moreover, bilateral agreements for activities in South Asia and Southeast Asia would be beneficial in complementing current efforts being made under AoW 5. The most evident gap in financial complementarity appears to be in AoW 3 (policy), which relies exclusively on pooled funding, and in the total financial coverage needed to expand AoW 2 (user engagement).

It must be acknowledged, however, that most of the work done under AoW 3 is designed to service CGIAR's own direct interests: i.e., a) ensuring CGIAR complies with laws related to generic resources, digital sequence information and farmers' rights (so that CGIAR does not end up suffering financial or reputational loss as a result of non-compliance), and b) contributing to development of future policies that will support (and not hinder) CGIAR's mission. That said, it is also important to note that national partners with whom AoW 3 shares capacity to develop national policies make contributions in kind that are much larger than the relatively small financial support they receive from AoW 3.

Strategic use of bilateral and external funds helps close gaps, but more predictable, adequate core funding is essential to fully realize the Accelerator's global vision and address stakeholder priorities comprehensively. Pooled funding provides the essential backbone for all AoWs, ensuring fundamental operations and crucial policy work.

The bilateral mapping exercise proved to be complex for the genebanks as the Accelerator had been initially listed as a Program and merged with Breeding for Tomorrow.

Changes are still being implemented to the mapping exercise, the results are accurate at the time of drafting. Information may differ on the Bilaterals dashboard after this date. Further updates are envisioned during Phase 2 of the Mapping exercise. The below list was requested by the Program Coordination Unit given the special case of missing bilaterals for the Genebanks.

The following projects were initially listed as Genebanks, after the Accelerator was split from Breeding:

- D-100735-Barley quality traits

- GCDT-Long-term conservation and sustainable use of plant genetic resources (Alliance Bioversity-CIAT)
- 1572-MIPO Impulsando la Agroindustria Rural y el Uso de la Diversidad del Yacón conservada en Pataz y CIP, a través de la Bioeconomía y los Alimentos Funcionales
- P-1560-DGUL-Andean Crop Diversity for Climate Change
- 1571-FAO0 - Youth, Citizen Science and E-commerce: scaling integrated conservation solutions and farmers' rights by connecting key diversity hotspots: Bolivia, Chile, and Peru
- P-1530-GIZ0-Construction and implementation of a new Cryobank
- Conservation of ICRISAT genetic resources for food and nutrition security in the semi-arid tropics

The following Genebanks projects were erroneously listed as belonging to other Science Programs/Accelerators:

- 1572-MIPO (CIP) Impulsando la Agroindustria Rural y el Uso de la Diversidad del Yacón conservada en Pataz y CIP, a través de la Bioeconomía y los Alimentos Funcionales (Better Diets and Nutrition)
- P-1560-DGUL (CIP)-Andean Crop Diversity for Climate Change (Better Diets and Nutrition)
- Conservation of ICRISAT genetic resources for food and nutrition security in the semi-arid tropics (Breeding for Tomorrow)
- T-PJ-003670- Long-term funding of Ex Situ collections of germplasm (IITA) (Breeding for Tomorrow and CapSha)
- L-LTG001-Long term funding of ex situ collections of germplasm (ILRI) (Breeding for Tomorrow)
- R-A-2023-9-Genebank-AI (Breeding for Tomorrow)
- T-PJ-003575: Cassava Allele Mining Project: (IITA) Mining useful alleles for climate change adaptation in the cassava from CGIAR Genebanks (Breeding for Tomorrow and Climate Action)
- T-PJ-003576- (IITA) Mining useful alleles for climate change adaptation cowpea (Cowpea Allele mining project: Mining useful alleles for climate change adaptation in cowpea from CGIAR Genebanks (year 2022-26) (Breeding for Tomorrow, CapSha, and Climate Action)
- T-PJ-004017-VACS (IITA): Vision for Adapted Crops and Soil: Bambara Groundnut (Breeding for Tomorrow, CapSha and Better Diets and Nutrition)
- T-PJ-004023-VACS (IITA)-Breeding: Vision for Adapted Crops and Soil: Taro (Breeding for Tomorrow, CapSha and Better Diets and Nutrition)
- Regeneration of maize accessions from the Highlands of the State of Mexico for the ICAMEX's Germplasm Bank (Multifunctional Landscapes)
- R-A-2024-3-2024-2028LPA Long-term Partnership Agreement (IRRI) (Breeding for Tomorrow)

There was a set of bilateral projects related to the work of the genebanks which were not initially selected among the list of bilaterals as per the CGIAR Centers' methodology, but they are nevertheless important for the core functions of the genebanks as well as to track relevant indicators associated with the Accelerator's theory of change.

- Long Term Partnership Grant for conservation of the rice biodiversity at AfricaRice genebank (AfricaRice)
- Long Term Partnership Grant (CIP)
- Fortalecimiento de la conservación y uso sostenible de variedades locales de raíces y tubérculos andinos libres de enfermedades en la Zona de Agrobiodiversidad Andenes de Cuyocuyo- Puno (CIP)
- Long Term Partnership Agreement (ICARDA)
- Providing for the long-term funding of ex situ collections of germplasm (Bioversity)
- Long Term Partnership Grant (ICRISAT)
- Defining new phenotypes for forage and crop residue improvement based on rumen function and greenhouse gas emissions (UK-CGIAR)

Finally, there are projects which are crucial to the work of Genebanks but were either not assigned to a Science Program/Accelerator during the mapping exercise, or that have no budget assigned to the Genebanks but are still relevant in terms of work and objectives for the Genebanks team.

- Global Crop Diversity Trust (GCDT)-LTG Support to Wheat and Maize Genebank (CIMMYT)
- T-PJ-004010: Enhancing production of locally adapted, climate resilient underutilized crops (IITA)
- Screening, developing, and deploying anti-methanogenic feedstock into livestock systems in the Global South BIUSA (BEF)
- Screening, developing, and deploying anti-methanogenic feedstock into livestock systems in the Global South
- Anti-methanogenic feedstock for livestock systems in Global Systems

7. Plan of Results and Budget

The Genebanks Accelerator's 2025 Plan of Results and Budget (PORB) is grounded in CGIAR's legal and strategic mandate to conserve and make available crop diversity through a globally coordinated system of international genebanks. The PORB is structured to ensure the delivery of high-level outputs across the five AoWs, with a strong emphasis on continuity and operational integrity.

The PORB reflects the foundational role of AoW 1 (Biodiversity Conservation), which receives the largest share of W1/2 pooled funds. This allocation safeguards the core functions of seed and clonal genebank operations, which are non-discretionary and legally mandated under the Plant Treaty. AoW 1 also benefits from stable long-term partnership grants and agreements managed through the Crop Trust endowment.

AoWs 3 (Policy) and 4 (Germplasm Health) are also prioritized, reflecting their critical roles in ensuring ABS and phytosanitary regulation compliance, involvement in key policy proceedings, and enabling global germplasm exchange.

AoWs 2 (User Engagement) and 5 (Strengthening capacity), though essential for long-term impact and outreach, are funded at a lower level under current W1/2 allocations. Nevertheless, the impact of bilateral and W3 projects is extremely beneficial in supporting these AoWs.

Most recently, in response to a donor earmarking US\$1.26 million specifically for AoW 5—a significant increase from the originally budgeted US\$ 530,000—the Genebanks Accelerator has reorganized existing W1/2 funds to match the donor contribution. This US\$ 730,000 adjustment was not a result of new funding but was reallocated from other AoWs to align with the updated funding envelope under AoW 5. The reallocation affected multiple Centers:

- Bioversity reallocated US\$ 12,000 from AoW1 and US\$ 200,000 from AoW3 to AoW5.
- AfricaRice redirected US\$ 14,000 from AoW1 to AoW5.
- CIAT shifted US\$ 62,000 from AoW1 to AoW5.
- CIMMYT, ILRI, ICARDA, and IRRI each moved US\$ 12,000 from AoW1 to AoW5.
- IITA reallocated US\$ 132,000 from AoW1 and US\$ 50,000 from AoW4 to AoW5. 120K from AoW1 was transferred to CIP for cross-cutting activities to be implemented in LAC.
- CIP shifted US\$ 212,000 from AoW1 to AoW5.

This strategic redistribution ensures full use of the earmarked funds and supports a broader and more impactful delivery of AoW5 activities in 2025.

Budget allocations in the PORB reflect the operational scale, risk exposure, and current performance of each Center's genebank. Centers hosting clonal crops receive proportionally higher budgets due to the more resource-intensive nature of their conservation operations, including cryopreservation.

The PORB integrates findings from the comparative advantage and prioritization exercises conducted during the Inception Phase. It preserves core activities in AoWs 1, 3, and 4 while enabling incremental growth in AoWs 2 and 5 based on emerging needs and available bilateral support.

While the Genebanks Accelerator sustains long-standing operations, the 2025 PORB also reflects adaptive and forward-looking strategies:

- Investment in **AI-enabled tools** to enhance precise germplasm requests and trait discovery
- Expansion of **regional partnerships**, considering the results of the prioritization exercise and experience with partners

- Greater alignment with **CGIAR programs** like Breeding for Tomorrow, Food Frontiers and Security, Digital Transformation, and Capacity Sharing, with joint workstreams integrated into HLOs under AoW 2 and 5.

8. Cross-Portfolio linkages and geographic coordination

8.1 Cross-portfolio linkages

The main cross-portfolio linkages are:

Breeding for Tomorrow (B4T): Having a successful and two-way partnership with B4T is relevant to all Areas of Work in particular AoW2, HLO 2.2 *Enhanced germplasm information for more precise accession targeting*. The reason for this is that breeding analysis can produce interesting data to feed the genebanks' databases and, in turn, enhance accession targeting and diversity use. However, some distance should also be maintained to avoid the genebanks evolving into working collections of the breeding programmes. Nevertheless, for this strategy to succeed, the partnership must be mutually beneficial, genebanks must make material available and support breeders in the selection process and breeders must provide post-analysis data and feedback on the accessions to inform genebanks' databases. This process was initially implemented by including pre-breeding roles in some of the CGIAR genebanks; however, these roles were discontinued due to lack of funding and also for being considered a deviation from the core role of the genebanks. The Genebanks Team believe initiatives that promote a better linkage with breeding should be implemented to operationalize this partnership with B4T. The new programmes of breakthrough products of BMGF could also be an interesting platform to discuss a refresh of the collaboration between B4T and Genebanks. As per the ISDC recommendations "activities including partnerships between genebanks and breeding to work with partners to preserve endangered diversity (including Neglected Underutilized Species) (Areas of Work 2 and 5) are vitally important to address the climate and biodiversity crises." For AoW 4, B4T should support exchange of germplasm for breeding trials and seed systems.

Digital Transformation: the Digital Transformation Accelerator is already providing financial and technical support on a project with AWS hosted at CIP and is also part of the collaboration with Google that finances a project at IRRI. Further collaboration is envisioned especially as related to HLO 2.3 *CGIAR-wide portal with AI-enabled digital infrastructure developed for germplasm data and requests* and 2.4 *Genebank products developed to enhance diversity use*. Both outputs require the use of AI to enhance broad-base characterization of accessions and AI-supported subset generation.

Sustainable Farming: GHUs under AoW4, will collaborate with Sustainable Farming Program AoW4 (Plant Health and Mycotoxin Safe Crops) in the areas of diagnostics and surveillance of invasive pests and diseases, support national diagnostic networks and facilitate exchange of non-plant taxa materials.

Multifunctional Landscapes: the work of the Genebanks, particularly under AoW1 and AoW5, strongly aligns with the priorities of Multifunctional Landscapes Science Program Area of Work 1, that builds on an understanding of landscapes as complex and interlinked social-ecological systems, with biodiversity conservation as one of the core elements of these systems.

Food Frontiers and Security: In response to an earmark funding to AoW4 Island Food Systems, this program will be working in collaboration with the Genebanks Accelerator on theme 1: Documenting, managing and using genetic diversity in the Pacific Islands. This is positively seen as an opportunity to start strengthening the collaboration between the two programs.

Finally, the **Capacity Sharing Accelerator** could support connecting capacity development needs with available expertise, centralizing opportunities and measuring and operationalizing the impact of the genebanks' capacity development products. This is particularly relevant to all HLOs in AoW5. During the Genebanks Inception Meeting in May 2025 we engaged with CapSha focal point

for Genebanks and started interesting discussions about the potential engagement with this accelerator.

During the inception phase the Genebanks Accelerator has attended the meetings of two Accelerators, Scaling and Digital Transformation to foster better and deeper collaboration.

8.2 Geographic coordination

B4T: currently the linkage is strong in a few Centers and across a few collections, but it should be better operationalized. This partnership has proven to be extremely effective in centers where breeding and genebanks collaborate through joint bilateral projects and initiatives that promote the better use of the germplasm in the benefit of the breeding programs, leading to the so needed mutual benefits. This level of partnership has the potential to be successful in all CGIAR genebanks because breeders maintain deep collaboration with the genebank resources area and the genebank team can use breeders' knowledge to optimize the use of those resources. In most Centers, genebank staff have informal connections with their breeding counterparts. Currently, breeding programs deposit released lines and genetic stocks in genebanks. Ideally, the genebanks would have a more formalized relationship with breeding globally, including regular check-ins on the use of accessions and a stronger partnership on user engagement, as also suggested by the ISDC recommendations (see section 8.1 of this document). There are currently a few bilateral projects where B4T and genebanks share funding and the genebanks intend to continue to draw on these collaborations (for more details on these, consult the W3 bilateral funding dashboard and the PORB).

Digital Transformation: The Digital Transformation Accelerator's partnership with the Abu Dhabi government on the use of AI and new technologies for agriculture could be of interest to the Genebanks Accelerator to increase the use of AI for genebank operations. This will be coordinated mostly out of the IRRI and CIP genebanks but with global implications. Currently, the genebanks are interacting with Digital Transformation in collaboration with Google and AWS. This Accelerator will serve as a funnel for the genebanks to access funding and knowledge for new technologies and technical skills to implement these technologies.

Capacity Sharing: the genebanks plan to increase cooperation with Capacity Sharing over the course of the Accelerator. The two Accelerators have nominated focal points to increase communication globally. More partnerships are envisaged particularly in relation to training standardization, advice from Capacity Sharing on avoiding duplication globally and fundraising for training. Coordination with Capacity Sharing will be done regionally through our offices located in all crop diversity hotspots. AoW5 is led out of AfricaRice (Côte d'Ivoire) and CIMMYT (Mexico), where the CapSha focal point for the Genebanks Accelerator is also located.

9. Risk management

Please refer to the following [link](#) with the Accelerator's full risk register. This link is open to ISDC colleagues and others will be granted access, on request.

#	Category	Title	Description	Current Risk Level	Target Risk Level	Actions /Controls to manage risks
1	Data	Partners and users data sharing and collaboration failure	CGIAR Centers unable to recover information generated by requestors on the use of the CGIAR genebanks materials, due to recipients data sharing restriction policies or unwillingness to share data. This impacts CGIAR Genebanks ability to progress on the development of unified approaches for data management infrastructure, harmonized processes and the development of a portal to facilitate users' access to genebank data and germplasm ordering. This could lead to research not advancing at the pace envisaged and operational inefficiencies.	12	12	Raise awareness on the importance of sharing data among Genebank Managers
2	Business continuity	Operational disruptions from management changes	Changes of management mechanisms, coordination team and budget cause disruptions in genebank operations,	12	12	

			program activities or changes the objectives of the program. This is particularly relevant in years of budget uncertainties, when the Genebank Accelerator can be affected by lack of funds across the portfolio, and if not prioritized, with the ringfenced mechanism being acknowledged and respected, can deeply affect the continuity of the operations.			
3	Business continuity	Policy barriers to germplasm exchange	The current International Policy Framework (CBD, Plant Treaty, Nagoya protocol) changes, or ceases to exist in the current model and undermines the ability of CGIAR Genebanks to share germplasm with national genebanks and other users, restricting our ability to partner with outside organizations in activities that involve exchanging germplasm and digital sequence information,	9	9	Advocate for the importance of sharing germplasm and information across genebanks at these international fora

			resulting in the unavailability of biodiversity to address program objectives and outcomes.			
4	Business continuity	Unrest and emergencies impacting genebank operations	Unrest or unpredictable events occur in countries hosting Genebanks. This includes the emergence of new pests and diseases, as well as natural disasters or human made impacts such as conflicts, unrests, and wars. These diminish the capacity of the team to achieve the intended outputs of the program and demand an immediate reallocation of resources for recovery actions, potentially putting in risk the personnel and the germplasm.	9	9	Set up of contingency fund to address these emergencies without reallocating budget
5	Funding	Insufficient capacity and misuse of resources	There is insufficient capacity, resources or authority of management structures to facilitate the coordination of collective work under the Areas of Work. This also applies to the technical and	4	4	

			financial oversight of the program as a whole. A second element to this risk is the misuse of funds for purposes outside what is set up in the Workplans of the Areas of Work, or outside the Genebanks Accelerator objectives.			
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10. Addressing feedback on selects topics

Concise summary of feedback	Details on how the Program/ Accelerator has addressed or will address this feedback (in the Inception Phase or beyond)
There is a lack of information on how water systems are covered in the Portfolio.	N/A
There is a lack of information on how climate mitigation is covered in the Portfolio.	N/A
There is a lack of information on how tradeoffs and synergies in mixed systems are covered in the Portfolio.	N/A
There is a lack of information on how environmental health and biodiversity is covered in the Portfolio.	The Genebanks proposal noted that activities will lead to a stronger overall global system of international and national institutions, policies, and mechanisms contributing to the conservation of unique plant genetic resources, responding to SDG 2.5 and the targets of the Kunming-Montreal Global Biodiversity Framework, and giving hundreds of thousands of new user's access to plant diversity sources. AoW 4, specifically output 4.4, allows prioritization of germplasm health through phytosanitary measures. This can bring innovations in pest detection and treatment to safeguard genetic resources, supporting global food security and resilience against climate change threats by directly impacting the health of plant genetic material that is shared globally.
There is a lack of information on how orphan and opportunity crops and pulses are covered in the Portfolio.	CGIAR genebanks conserve crops as per their Centers' mandates. Many of the genebanks conserve orphan crops, e.g. ICRISAT conserves small millets and CIP conserves Andean Roots and Tuber crops. Adding new crops represents increased costs and requires Center Boards approvals. This is one of the reasons why the PGRFA Global System and the capacity building provided by CGIAR is so important, because they also promote better conservation of other crops that are not within the mandate of CGIAR but are important to countries which have them in their collections.
There is a lack of information on how youth and social inclusion (that does not relate to gender) is covered in the Portfolio.	CGIAR genebanks distribute materials directly to farmers, individuals, and communities on a small scale. CGIAR genebanks distribute a few thousand samples to farming communities, representing around 8% of their total annual germplasm distributions (although this figure appears to be increasing). Certain crops, varieties, and traits (e.g., clonal crops such as potato and banana, forages, trees spp.) follow direct distribution routes more frequently than other crops depending also on the outreach of

Concise summary of feedback	Details on how the Program/ Accelerator has addressed or will address this feedback (in the Inception Phase or beyond)
	genebanks to different user typologies. Additionally, providing mixed varieties and crops in the form of seed kits can help respond broadly to communities' aspirations to diversify their agricultural outputs and landscapes and integrate multiple food groups for system-based solutions and home gardens. By acting upon trends in these direct germplasm distributions, genebanks may be able to contribute directly to social inclusion impact. AoW 5 (Strengthening capacity) provides an opportunity to change this lack of information and promote approaches to specifically target in situ and ex situ conservation and the two-way flow of germplasm, including monitoring of in situ sites. This AoW will also help to build capacity and provide safety backups to support in situ conservation of crops of mutual interest. Several genebanks nurture existing in situ – ex situ connections with communities where there remains a strong cultural connection to crop diversity and traditional knowledge.
There is a lack of information on how capacity sharing (as it relates to internal learning and decolonization of research) is covered in the Portfolio.	All capacity sharing promoted as part of the Genebanks Accelerator aims to support national genebanks and partners in implementing best practices and becoming more efficient in managing their collections and thus contributing to the conservation and availability of PGR . The in situ-ex situ collaboration follows the same path, supporting farmers in improving the in-situ conservation of their crops, making them more self-sustainable and independent.

Appendix 1: Addressing specific feedback

Concise summary of feedback	Details on how the Program/Accelerator has or will address this feedback (in the Inception Phase or beyond)
Lacks clarity on co-design, particularly in defining roles and responsibilities among key partners.	The proposal was developed together with the WorldVeg and CIFOR-ICRAF. The Crop Trust, Plant Treaty and national partners were also involved in consultation and provided extensive feedback. The template (for a research program) was not ideal to describe the largest part of the work that is involved in the Genebanks proposal, which is the routine operations of the genebanks, cryobanking, work to improve protocols, policy compliance and phytosanitary health. For more than 80 percent of the proposal, there is no co-design element since it relates to CGIAR genebanks' internal operations. AoW2 is designed to engage users to improve outreach and distribution of germplasm. This has already involved and will continue to involve codesign as the strategy is developed and implemented. AoW5 on capacity building with NARS by its nature depends entirely on developing activities in collaboration with national partners through regional entry points, and the indicators and goals have been developed with national partners' needs in mind (gathered through surveys, stakeholder consultations, in the context of training programs and based on FAO's State of the World Report on PGRFA and in country reports to the UN Commission on Genetic Resources for Food and Agriculture's Global Plan of Action for Plant Genetic Resource). (See more in section 2 of this document)
Section on MELIA fails to include essential learning questions that would guide reflection and inform adaptive interventions. Establishing baseline information for various activities is critical for meaningful impact assessment. It becomes challenging to measure progress.	The genebanks monitoring system is well established in partnership with the Crop Trust and has been functioning well since 2014. Baselines were developed at that date and progress towards performance targets has been clearly shown and long-term grants have been awarded as a result. The genebanks are not research programs and do not implement interventions in the conventional sense.
The focus should shift from increasing user numbers and seed requests to prioritizing core activities related to collection, conservation and distribution of genetic resources. CGIAR must identify regions where their interventions are truly needed. By emphasizing local capacity building and minimizing competition with national research efforts, CGIAR can foster sustainable agricultural development.	There is some paradox here since the reviewer mentions the focus should be related to "collection, conservation and distribution". Monitoring the number of users does relate to distribution. Alternative metrics for the numbers of samples and accessions distributed are already monitored but have not been imposed as a performance target since there is no need to incentivize sending out more germplasm when sending out more precisely matched and potentially fewer samples is preferable. Regarding the second point of the comment, it is important to note that CGIAR genebanks' work is all about complementarity with national partners. CG genebanks can play a role in expanding crop diversity conservation only through complementarity and coordination with NARS. Conservation activities are a major focus of the genebanks and are already well monitored through performance targets, audits and reviews. Again, it should be emphasized that the genebanks are not planning interventions. Germplasm is distributed from CGIAR genebanks upon request from the user. CGIAR genebanks do not send out materials unsolicited. However, we do acknowledge that the distribution can be improved in terms of interaction

	with users, better data mining and consequently more targeted materials can be selected to better fulfill the needs of the users. Regions where interventions are truly needed have been identified in the Inception Phase thanks to the Prioritization Exercise, more information can be found in Section 5 of this document.
Identify and mobilize key stakeholders and clarify their roles and responsibilities	Please note key stakeholders have been identified for each AoW (see Section 3.1). Key stakeholders and partners have also been involved in the co-design of specific activities, outputs and outcomes (see Section 2).
CGIAR's unique role in genetic resources needs more clarity. Additionally, the claim that CGIAR holds a unique position in providing genetic resources is questionable, given the presence of well-managed local and national genebanks. The proposal should clarify the uniqueness among genetic resources.	This feedback has mostly been addressed in the Comparative Advantage section of this document (Section 4). To further illustrate the comparative advantage of CG genebanks, it is important to note that they have benefitted from stable funding for 12 years which has impacted significantly the status of the collections and capacity to operate. This is not true for most of the national genebanks, especially in the global south. These findings were endorsed in two external reviews in 2017 and 2023. The concern for the need to regenerate ageing collections, safety duplicate them and provide data from national genebanks is acknowledged in FAO's State of the World Report on PGRFA and in country reports to the UN Commission on Genetic Resources for Food and Agriculture's Global Plan of Action for Plant Genetic Resources. These collections and genebanks are extremely vulnerable. Moreover, the unique position in providing genetic resources is validated by the high percentage of germplasm being shared by the CGIAR alone under the MLS of the Plant Treaty.
Details on cooperation and outreach are needed	It is useful to distinguish cooperation with partners from outreach to users. AoW 2 describes in detail the plans to develop strategic tools to facilitate user interaction with genebanks data optimizing the process of identification of germplasm suitable for the interests of the pretended use and facilitating the request. Cooperation with key partners such as the Plant Treaty, the Crop Trust, FAO Commission and with CGIAR partners is described for each area of work, as well as in the codesign section of this document (section 2).
An analysis of operational costs should be conducted	Operational costs were analysed in depth in 2010 and 2020 in a joint initiative with the Crop Trust. These previous studies are used as a baseline for updating operational costs in 2024 for use in the workplan and budget for 2025-2030. In the future, the Crop Trust will trigger another costing study, probably in 2030. The 2023 IAES external expert panel focused on genebank costs and reported that there was evidence of cost savings since 2011. It is unlikely that any CGIAR operation or service has undergone such detailed scrutiny and technical review as the genebanks operations thanks to the oversight of the Crop Trust given their responsibility in managing an endowment mechanism to support these collections. The findings of a system-wide study of costs undertaken by the Crop Trust in 2020, highlighted that cost reductions should be focused on root, tuber and banana crops, forages and trees, which are comparatively expensive. Efforts to rationalize roles and responsibilities for these crops across Centers should be considered. In addition, rather than focusing on processes, which are already well developed and efficient, cost-efficiencies should be sought by providing greater transparency and effectiveness in how Center chargebacks are implemented, reducing non-attributable research support chargebacks that are included as a direct cost, and providing full control of the budget and retainment of staff to the genebank managers to enable and incentivize rationalization.

Comparative advantage analysis of GHU is weak.	GHUs were established to meet mandatory phytosanitary requirements for international germplasm exchange by CGIAR programs in response to the absence of suitable service providers capable of fulfilling this mandate. A comparative advantage analysis of GHUs would have to consider that their phytosanitary service work is intertwined with the CGIAR genebanks and breeding program activities concerning acquisitions, seed regeneration, conservation, and distribution. These services cannot be effectively undertaken by National Plant Protection Organizations (NPPOs), whose mandates are generally limited to import/export quarantine, nor by other external providers without being embedded in CGIAR operations and locations to handle large sample volumes efficiently. No comparable in-country provider currently offers this range of services at a lower cost or with reduced risk. For instance, a survey of NPPO capacity in the global south, especially the countries of CGIAR operations, organized by the Plant Health Initiative in 2022 revealed diagnostic capacity is inadequate for efficient phytosanitary monitoring. In addition, outsourcing phytosanitary controls would require local providers to avoid shipping untested samples internationally, which could create bottlenecks. The Musa Genebank at the University of Ghent, managed by the Alliance of Bioversity, has outsourced GHU functions to the University of Gembloux in Belgium. This model works because of the Musa genebank's distinct origin and setup (Musa transit center of INIBAP to A15 CGIAR genebank). The Gembloux GHU team is part of the CGIAR GHU network and follows the same operational framework. The GHU services at Gembloux are covered through the CGIAR Genebanks funding, with infrastructure and staff costs covered by the Belgian government as in-kind support. Before this arrangement, health testing of Musa germplasm was performed at the University of Queensland, Australia, due to dwindling financial support for this activity at the University of Queensland. Other CGIAR Centers lack both the necessity and opportunity to adopt a similar model. Outsourcing GHU services more broadly would likely increase costs and risks related to service continuity, turnaround time, and contractual complexity. Furthermore, such outsourcing would not eliminate the need for CGIAR to conduct phytosanitary activities; these would remain essential regardless of the delivery model. GHUs have co-evolved with CGIAR, delivering accounting change regulatory and pest scenarios, contributing to building phytosanitary capacity in national programs across the Global South. The GHUs serve as diagnostics hubs in the region, supporting a wide range of national and regional surveillance activities. GHU model is acknowledged as key to the success of CGIAR international germplasm exchange programs. For more details about GHUs: https://www.mdpi.com/2223-7747/10/2/328
Incorrect Statements that must be addressed: The claim that CGIAR holds a unique position in providing genetic resources is questionable, given the presence of well-managed local and national genebanks.	CG genebanks' role is not to conserve all crops diversity, but to enable national genebanks to efficiently and effectively conserve the broad genetic diversity they house (see AOW5). Even though almost 90% of the germplasm shared globally through the Plant Treaty's multilateral system of access and benefit sharing is from CGIAR Centers. All other providers –including all genebanks and breeding programs in the world– collectively only provide 11 percent of material exchanged under the MLS. Major contributing factors to the disproportionately high use of CGIAR-hosted ex situ collections are the genetic diversity of our collections, the fact that they are well documented, and the high level of confidence among recipients that our materials are clean of pests and diseases. Moreover, the fact that almost all accessions are immediately available to be distributed, not requiring multiplication, viability testing or other checks to trigger distribution. Looking at the data

	from the Plant Treaty, well managed genebanks are in the global north, with the USDA one providing the most germplasm.
Enhance focus on climate impacts and NUS, clarify CGIAR's global role, define how gaps will be identified, and critically evaluate the rationale behind increasing collection use. Define specific research questions to guide the proposal.	Enhancing the focus on climate impacts is very relevant. One of the ways genebanks can do this is by associating climate-relevant data (e.g. georeference) with the accessions to aid their data-driven selection by users. This is one of the objectives of AoW 2 as illustrated in section 3 of this document. Nevertheless, genebanks do not conduct direct research and are not aiming to carry out climate-related interventions but are ready to contribute with the Climate Action Science Program. CGIAR genebanks' role is to conserve and make available collections under their responsibility as designated under Article 15 of the Plant Treaty. Some NUS are conserved in our collections, especially those of the same group as major staples (e.g. Vigna, Phaseolus, small millets, etc). The involvement of WorldVeg would have considerably expanded the scope of the genebanks' work by including 500 species of vegetables including many NUS. Further genebanks could be involved if funding were available (e.g. CePaCT in Fiji manages the global collection of taros). Adding new crops to the CGIAR genebanks requires extra funds and more importantly Center Boards' approval. Numerous gap analyses of collections in CGIAR genebanks have been published using geographic, taxonomic and trait data, and are used for directing collecting efforts where funds are available for such activities. Working with national partners to jointly analyse genetic diversity of shared crops and identify priorities to build on the information already available from published studies and Global Crop Conservation Strategies published by the Crop Trust is part of the foreseen activities for Area of Work 5.
Theory of change would benefit from better defined impact pathways and SMART goals	See section 3 of this document and below.
Clearly define the impact path and simplify the presentation of interactions, potentially by reorganizing sections. Establish SMART goals for accountability. Provide more clarity on the role of genebanks in breeding and avoid overstating the benefits of virus-free materials without addressing multiplication challenges.	There is no predetermined impact pathway in this instance – a reminder that genebanks are considered a service. They do not plan interventions or specific outcomes. The proposal attempted to describe the range of different potential impact pathways depending on the crop, region and the relative resourcing of the crop breeding pipeline. As a service, it is more appropriate for genebanks to measure whether we are reaching users who can benefit from genebank services and whether those users are satisfied. There is a grave misunderstanding implied in the recommendation to “avoid overstating the benefits of virus-free materials without addressing multiplication challenges”. The genebanks are not suppliers of source seed for seed systems, although this may be requested of them especially for root, tuber and banana crops. They provide small amounts of germplasm of a wide range of accessions representing available crop diversity for a wide range of largely research and evaluation purposes, frequently to users from other countries. It is true that germplasm generally does need to be multiplied up in-country. However, if the germplasm from CGIAR arrives clean then in-country multiplication can only propagate existing in-country pathogens. This is true also for the source seed that CGIAR provides from breeding programs, which are also serviced by GHUs. CGIAR's reputation would be seriously exposed if it were responsible for the spread of pests and diseases from one country to another or worse from one continent to another. The reason that CGIAR genebanks and breeding programs have high levels of international distribution is because researchers can be assured that the materials they request are not infected by pathogens. There

	are few alternative sources, especially for roots, tubers and bananas, that can provide the same assurance.
AoW1 Emphasize the inclusion of NUS in Genebank collections and clarify measurement criteria for process improvements. Identify specific bottlenecks and establish accountability for targets, along with clear baselines. Provide rationale for chosen technologies and detail how various platforms integrate to enhance effectiveness. Set funding milestones to ensure support for additional activities.	See above for response on NUS. Note also that CGIAR genebanks will always be responsible for the conservation and availability of the major staple crops, mandate crops of the Centers that host the genebank, and that resources cannot necessarily be moved from these to other crops especially given the frequency of requests for these crops. Hundreds of small genebank process adjustments and improvements are made annually. These are extremely hard to monitor and document but a better job should be done to enable us to communicate such improvements to funders. This includes how Videometer and other imaging efforts are having an impact on identifying and characterising accessions and helping to make genebank data AI-ready. AoW 1 illustrates in detail the rationale for certain technologies but this needs to be expanded including setting funding milestones which is currently ongoing.
AoW2 Enhance user engagement by establishing a feedback loop with national-level stakeholders. Clearly define the impact of increased user participation and set specific targets and funding needs for Output 2.2. Assess the feasibility of adopting new technologies for local genebanks and emphasize the role of universities in training future professionals on germplasm use.	There is already a feedback loop with users: genebanks send out questionnaires (following distributions). DOIs are already established to trace use. Nevertheless, the goal of genebanks is not only to increase use but more importantly to improve interaction, reducing the interaction time for seed request, at the same time improving data and accession-selection processes.
AoW3 In futureproofing the ABS policies, the impact of climate change will be largely overlooked. Most of the conservation efforts are made at the national (local level) with limited resources and capacity to climate change impacts (adaptation), especially in low-income countries. The benefit sharing could play an important role by channelling resources to the conservation (maintenance) and use of crop genetic resources for adaptation purposes	Genebanks AoW 3 added a support activity: Capacity building to access climate adapted material (with AoW 2). National level policy development work will focus explicitly on identifying and accessing climate change adapted materials under the Plant Treaty. Genebanks AoW 3 reframed its objectives during the Inception Phase to highlight that the emphasis of the policy work on the ITPGRFA (bearing in mind that it is important to ensure that the ITPGRFA evolves and is implemented in mutually supportive ways with the CBD's Nagoya Protocol on ABS and Decision 16/2 on digital sequence information, because these two instruments also govern some important aspects of genebanks and breeders activities, as discussed below.).

especially where there are more adverse and urgent effects of climate change. The emphasis of the policy work should be on the International Treaty on Plant Genetic Resources for Food and Agriculture because this is the mandate of the CGIAR. The CBD is marginal for PGRFA. Finally, to be accountable for these activities the objectives need to be SMART.	
Incorrect Statements that must be addressed: Most of the conservation efforts are made at the national (local level) with limited resources and capacity to climate change impacts (adaptation), especially in low-income countries.	Fewer countries now account for a larger percentage of the total world ex situ germplasm holdings; many important collections are conserved inadequately hence potentially at risk. Many countries still lack adequate human capacity, facilities, funds or management systems to meet their ex-situ conservation needs and obligations, and as a result, several collections are at risk. Currently, although the Svalbard Global Seed Vault has more than 11 depositors of seeds, including almost 100 countries, around 50% of seeds safety duplicated there were sent by the CGIAR Genebanks, highlighting the efforts on implementing and sustaining high level standards on its operations and how far other genebanks still are to reach the CGIAR Genebanks capability. Greater efforts are needed to build a truly rational global system of ex situ collections. (see AOW5)
AoW5 Expand diversity analysis to include other crops and regions. Clarify how to effectively use limited resources and consider direct funding to regional genebanks. Define "low-cost genotyping" and evaluate the benefits of outsourcing this. Empower local scientists through targeted training and use of public platforms for broader outreach, ensuring regional governance develops independently without reliance on external intervention.	A training module on public platforms (e.g. YouTube) to appeal to a broader scientific public relating to conservation approaches is certainly appealing. However, there are other CGIAR programs and organizations outside the CGIAR who are better placed and who have already developed resources for community genebanks. CGIAR genebanks are ready to partner with those programs and organizations (e.g., Oxfam Novib), and have been doing some work related to community genebanks, as CIP Genebank does in Peru, and nature positive projects to provide relevant experience to help develop such widely relevant training tools. To quote the proposal "Building on a low-cost, joint initiative to genotype and share data on common bean diversity among national partners Latin America, further crops will be subject to similar analyses in Centers of crop diversity". This means that a low-cost effort was undertaken as a model and beans were mentioned as an example. This model is being used on other crops already and will now be expanded to more crops and countries, leveraging internal capacity and partnership among national partners. Outsourcing is not relevant here.
Incorrect Statements that must be addressed: The CBD is marginal for PGRFA.	Most collecting of new PGRFA to be introduced into CGIAR genebanks is governed by national measures implementing the Convention on Biological Diversity and its Nagoya Protocol. Recall that the genetic material within a country that are included in the Plant Treaty's multilateral system of access and benefit sharing (MLS) are PGRFA of the 64 crops and forages listed in Annex 1 of the Plant Treaty that are 'under the management and control' of the national government and 'in the public domain. In most countries, this

	formula is not interpreted to include PGRFA found in situ conditions or managed by farmers on farms. The result is that the collection of PGRFA from in situ and on farm conditions, to be included in Article 15 collections under the Plant Treaty, needs to comply with national measures implementing the CBD and its Nagoya Protocol. This is why mutually supportive implementation of the CBD/Nagoya Protocol, and the Plant Treaty are the subject of numerous resolutions by both the CBD's Conference of Parties and the Plant Treaty's Governing Body. Highly relevant discussions are ongoing under the CBD regarding PGRFA such as DSI, which requires deep involvement and participation of CGIAR, since it reflects on all the work that CGIAR (not only genebanks but breeders and other science areas) does on genotyping, bioinformatics and data availability.
Balance innovation with stability by ensuring new technologies enhance practical processes. Develop clear KPIs and SMART targets to evaluate progress and effectiveness, particularly regarding accessibility and distribution of cryo-conserved materials.	AoW 1 represents the sustained and improved operations of the genebanks. This area represents more than 70 percent of the proposed budget and activities. New technologies will be targeted exclusively to enhance existing processes and accession selection. Cryopreserved materials are designed as a long-term storage mechanism for the base collection. It is expensive (labour intensive) to introduce new materials into cryopreservation. Once it is stored in liquid nitrogen, its maintenance, however, is cheap and safe. It is, therefore, not the optimal storage mechanism meant for distribution and unlikely ever to be. Instead, the cryopreserved collection can act as the safety duplicate, and it can moreover facilitate the rationalization of the field and in vitro conservation forms. Those accessions that are not or hardly requested can be stored solely in cryopreservation. Its main purpose therefore is to vastly improve the long-term storage of roots, tubers, bananas and potentially other recalcitrant seed species and to facilitate the rationalization of overall genebank operations of these crops. More on these topics and their respective indicators can be found in the ToC section (AoWs 1 and 5).
Explore opportunities for partnerships with national research centres and advanced research institutes	We have been conducting regional workshops and promoting the implementation of regional networks with the objective to identify key partners on genetic resources conservation. Regional workshops were conducted in Asia and Africa, networks and CoPs were implemented in Latin America, and we plan to promote the continuity and scale of this work if an adequate budget is made available. The Annual Genebanks Meeting, which is organized each year in a different part of the world and includes NARS and other partners from the region and from the CGIAR Portfolio. Regional inclusivity is part of our focus on germplasm distribution, capacity building, networks and CoPs.